QUIZ

QUESTION1 (10 marks)

Find the eigenvalues of $A=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 4 & -17 & 8\end{bmatrix}$.

QUESTION 2 (10 marks)

Show that the following matrix is not diagonalizable:

$$A=\begin{bmatrix}1 & 0 & 0\\ 1 & 2 & 0\\ -3 & 5 & 2\end{bmatrix}$$

QUESTION 3 (10 marks)

(a) Show that vectors $\vec{w_1}=(0,2,0)$, $\vec{w_2}=(3,0,3)$, $\vec{w_3}=(-4,0,4)$ form an orthogonal basis for R^3 with the Euclidean inner product and use that basis to find an orthonormal basis by normalizing each vector.

(b) Express the vector $\vec{u}=(1,2,4)$ as a linear combination of the orthonormal basis vectors in part (a).

QUESTION 4 (20 marks)

Find the

(a) least squares solution

(b) least squares error

of the linear system $x_1-x_2=4$

$$3x_1+2x_2=1$$

$$-2x_1+4x_2=3$$

QUESTION 5 (10 marks)

Construct a spectral decomposition of the matrix $A=\begin{bmatrix}1 & 2\\ 2 & -2\end{bmatrix}$.